DK1 battery trading: methods

github.com/jacobazu/dk1-battery-trade · October 2026

A 2 MWh / 1 MW battery planned every day on my DK1 price forecast, with its risk, costs and P&L. Python, checked against the R version.

Plan	Mixed-integer linear program on the ensemble forecast, and on each model alone to compare: 90% round trip, one cycle a day (Bradbury et al., 2014). Solved with SciPy.
Risk	EWMA-filtered historical simulation of the forecast errors (Hull and White, 1998; J.P. Morgan/Reuters, 1996), a 5% floor and the average below it, the Kupiec test (Kupiec, 1995), and four stress tests.
Loss limit	A limit on the worst 5% of scenarios (Rockafellar and Uryasev, 2000). Tested and not used.
Costs	Nord Pool fee and an assumed wear cost in the plan; Energinet's tariffs as a side case.
Tests	Daily differences tested with a Newey–West variance (Newey and West, 1987); Holm for the three models (Holm, 1979).
Rules	The plan only uses the forecast saved before the auction. Limits, cost cases and tests are set before running.

References

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